Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/12/2006
PI Name:	Balakotaiah, Vemuri Ph.D.		
Project Title:	Fundamental Studies on Two-Phase Gas-Liquid Flows through Packed-Beds in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bala@uh.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-743-4318
Organization Name:	University of Houston		
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PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77204	**Congressional District:**	18
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	01/17/2002	**End Date:**	10/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
Contact Email:	brian.j.motil@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: received NCE to 10/31/2006, per N. Hall (2/06?)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	McCready, Mark (University of Notre Dame) Motil, Briam (NASA Glenn Research Center)		
Grant/Contract No.:	NAG3-2697		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In the typical operation of a packed-bed reactor, gas and liquid flow simultaneously through a fixed bed of solid particles. Depending on the application, the particles can be of various shapes and sizes and provide for intimate contact and high rates of transport between the phases needed to sustain chemical or biological reactions. The packing may also serve as either a catalyst or as a support for growing biological material. NASA has flown two of these packed-bed systems in a microgravity environment with limited or no success. The goal of this research is to develop models (with scale-up capability) needed for the design of the physicochemical equipment to carry out these unit operations in microgravity. New insight will also lead to improvements in normal gravity operations. The main objective of this task is to provide a quantitative prediction based on analysis and verified by experiments of gas-liquid flow in packed-beds. It will provide a fundamental understanding of cocurrent two-phase flow in packed-beds in microgravity. The information and modeling from this task will significantly contribute to the fundamental understanding of the interactions between the surface tension, inertia and viscous forces in the absence of gravity for packed bed reactors. The packed-bed reactor is a critical unit operation for a wide variety of physicochemical and biological technologies including regenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During this reporting period, one Ph.D. degree was awarded as a result of this research. The dissertation title was, "Gas-Liquid Two-Phase Flow Through Packed Bed Reactors in Microgravity." A new reduced gravity aircraft experimental rig was completed to test new void fraction sensors. Technical support was provided for the completion of a ground-based x-ray calibration facility (developed under an SBIR contract).
Bibliography Type:	Description: (Last Updated: 01/12/2016)
Articles in Peer-reviewed Journals	Motil B, Nahra H, Balakotaiah V. "Hydrodynamics of packed bed reactor in low gravity." SAE 2005 Transactions Journal of Aerospace. 2005 Jul:SAE Paper 2005-01-3035. http://dx.doi.org/10.4271/2005-01-3035 , Jul-2005
Dissertations and Theses	Motil BJ. "Gas-Liquid Two-Phase Flow through Packed Bed Reactors in Microgravity." Dissertation, Case Western Reserve University, Cleveland, Ohio, December 2005. , Dec-2005